

COSC 594

DM Alignment (III)

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Announcement

Check Canvas for Homework 2

Due on 10/10, 11:59pm

Outline

- RECAP

- INFERENCE-TIME ALIGNMENT

Outline

■ RECAP

Optimal policy

■ FINE-TUNING W/ PATH-WISE KL-REGULARIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \beta \mathbb{E}_{x_{0:T} \sim p(\cdot)} \left[\sum_{t=T+1}^1 D_{\text{KL}}(p_t(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t)) \right]$$

$\beta > 0$ – tradeoff parameter

★ optimal policy $p^*(x_{0:T}) = \prod_{t=T}^1 p_t^*(x_{t-1} | x_t) p_{T+1}(x_T)$

$$p_t^*(x_{t-1} | x_t) = \underset{p_t}{\text{argmax}} \quad \mathbb{E}_{x_{t-1} \sim p_t(\cdot | x_t), x_{t-2} \sim p_{t-1}^*(\cdot | x_{t-1}), \dots, x_0 \sim p_1^*(\cdot | x_1)} \left[r(x_0) - \beta \sum_{k=t-1}^1 D_{\text{KL}}(p_k^*(\cdot | x_k) \| p_k^{\text{pre}}(\cdot | x_k)) \right] - \beta D_{\text{KL}}(p_t(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t))$$

■ OPTIMAL VALUE FUNCTION

$$v_{t-1}(x_{t-1}) = \mathbb{E}_{x_{t-2} \sim p_{t-1}^*(\cdot | x_{t-1}), \dots, x_0 \sim p_1^*(\cdot | x_1)} \left[r(x_0) - \beta \sum_{k=t-1}^1 D_{\text{KL}}(p_k^*(\cdot | x_k) \| p_k^{\text{pre}}(\cdot | x_k)) \right]$$
$$v_0(x_0) = r(x_0)$$

■ STAGE-WISE POLICY OPTIMIZATION

$$p_t^*(x_{t-1} | x_t) = \underset{p_t}{\operatorname{argmax}} \mathbb{E}_{x_{t-1} \sim p_t(\cdot | x_t)} [v_{t-1}(x_{t-1})] - \beta D_{\text{KL}}(p_t(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t))$$

★ optimal policy at time t is $p_t^*(x_{t-1} | x_t)$

$$p_t^*(\cdot | x_t) \propto p_t^{\text{pre}}(\cdot | x_t) e^{v_{t-1}(\cdot)/\beta}$$

Expensive policy gradient $\Rightarrow$ **Online training-free sampling**

Challenges

Question #1: How to estimate an optimal value function $v_t(\cdot)$?

Question #2: How to sample from a normalized distribution?

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Question #1: How to estimate an optimal value function $v_t(\cdot)$?

Recursive optimal value function

$$\begin{aligned}
 v_t(x_t) &= \mathbb{E}_{x_{t-1} \sim p_t^*(\cdot | x_t), \dots, x_0 \sim p_1^*(\cdot | x_1)} \left[r(x_0) - \beta \sum_{k=t}^1 D_{\text{KL}}(p_k^*(\cdot | x_k) \| p_k^{\text{pre}}(\cdot | x_k)) \right] \\
 &= \mathbb{E}_{x_{t-1} \sim p_t^*(\cdot | x_t)} \mathbb{E}_{x_{t-2} \sim p_{t-1}^*(\cdot | x_{t-1}), \dots, x_0 \sim p_1^*(\cdot | x_1)} \left[r(x_0) \right. \\
 &\quad \left. - \beta \sum_{k=t-1}^1 D_{\text{KL}}(p_k^*(\cdot | x_k) \| p_k^{\text{pre}}(\cdot | x_k)) - \beta D_{\text{KL}}(p_t^*(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t)) \right] \\
 &= \mathbb{E}_{x_{t-1} \sim p_t^*(\cdot | x_t)} \left[v_{t-1}(x_{t-1}) - \beta D_{\text{KL}}(p_t^*(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t)) \right] \\
 &= \sup_{p_t} \mathbb{E}_{x_{t-1} \sim p_t(\cdot | x_t)} \left[v_{t-1}(x_{t-1}) - \beta D_{\text{KL}}(p_t(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t)) \right] \\
 &= \beta \log \mathbb{E}_{x_{t-1} \sim p_t^{\text{pre}}(\cdot | x_t)} \left[e^{v_{t-1}(x_{t-1})/\beta} \right] \\
 &\qquad\qquad\qquad p_t^*(\cdot | x_t) \propto p_t^{\text{pre}}(\cdot | x_t) e^{v_{t-1}(\cdot)/\beta} \\
 &\Rightarrow e^{v_t(x_t)/\beta} = \mathbb{E}_{x_{t-1} \sim p_t^{\text{pre}}(\cdot | x_t)} \left[e^{v_{t-1}(x_{t-1})/\beta} \right]
 \end{aligned}$$

Closed-form optimal value function

$$\begin{aligned}v_t(x_t) &= \beta \log \mathbb{E}_{x_{t-1} \sim p_t^{\text{pre}}(\cdot | x_t)} \left[e^{v_{t-1}(x_{t-1})/\beta} \right] \\&= \beta \log \mathbb{E}_{x_{t-1} \sim p_t^{\text{pre}}(\cdot | x_t)} \mathbb{E}_{x_{t-2} \sim p_{t-1}^{\text{pre}}(\cdot | x_{t-1})} \left[e^{v_{t-2}(x_{t-2})/\beta} \right] \\&\quad \dots \\&= \beta \log \mathbb{E}_{x_{t-1} \sim p_t^{\text{pre}}(\cdot | x_t)} \mathbb{E}_{x_{t-2} \sim p_{t-1}^{\text{pre}}(\cdot | x_{t-1})} \dots \mathbb{E}_{x_0 \sim p_1^{\text{pre}}(\cdot | x_1)} \left[e^{v_0(x_0)/\beta} \right] \\&= \beta \log \mathbb{E}_{x_0 \sim p^{\text{pre}}(\cdot | x_t)} \left[e^{r(x_0)/\beta} \right] \\&\approx \beta \log e^{\mathbb{E}_{x_0 \sim p^{\text{pre}}(\cdot | x_t)} [r(x_0)]/\beta} \\&= \mathbb{E}_{x_0 \sim p^{\text{pre}}(\cdot | x_t)} [r(x_0)]\end{aligned}$$

$$v_0(x_0) = r(x_0)$$

$$p^{\text{pre}}(x_0 | x_t) = \int p_t^{\text{pre}}(x_{t-1} | x_t) \dots p_1^{\text{pre}}(x_0 | x_1) dx_{t-1:1}$$

Estimation of optimal value function

Data set : $D = \{x_{0:T}^{(i)}, r(x_0^{(i)})\}_{i=1}^N$

Value function class : $\phi(x, t): X \times [0, T] \rightarrow \mathbb{R}$

■ NONLINEAR REGRESSION

$$\underset{\phi}{\text{minimize}} \sum_{i=1}^N \sum_{t=T}^0 \left(v_t(x_t^{(i)}) - \phi(x_t^{(i)}, t) \right)^2$$

$$v_t(x_t^{(i)}) \approx r(x_0^{(i)})$$

$\Downarrow$

$$\underset{\phi}{\text{minimize}} \sum_{i=1}^N \sum_{t=T}^0 \left(r(x_0^{(i)}) - \phi(x_t^{(i)}, t) \right)^2$$

Approximate value function $\hat{v}_t(x_t) = \phi^*(x_t, t)$

Question #2: How to sample from a normalized distribution?

Importance sampling

$$p^*(x) = \underbrace{p^{\text{pre}}(x)}_{\text{easy}} \underbrace{\frac{e^{r(x)}}{\int p^{\text{pre}}(x') e^{r(x')} dx'}}_{\text{unknown}}$$

■ SAMPLES FROM PRETRAINED MODEL p^{pre}

$$x^{(1)}, x^{(2)}, \dots, x^{(M)} \sim p^{\text{pre}}(\cdot)$$

■ CATEGORICAL WEIGHTS

$$w^{(m)} = e^{r(x^{(m)})} \Rightarrow \frac{w^{(m)}}{\sum_{j=1}^M w^{(j)}} \text{ for } m = 1, 2, \dots, M$$

■ CATEGORICAL SAMPLING

$$x^{(\zeta)} : \quad \zeta \sim \text{Categorical} \left(\left\{ \frac{w^{(m)}}{\sum_{j=1}^M w^{(j)}} \right\}_{m=1}^M \right)$$

Empirical approximation

$$p^{\star}(x) = p^{\text{pre}}(x) \frac{e^{r(x)}}{\int p^{\text{pre}}(x') e^{r(x')} dx'} \approx \sum_{m=1}^M \delta_{x^{(m)}} \frac{w^{(m)}}{\sum_{j=1}^M w^{(j)}}$$

Larger $M \Rightarrow$ closer to $p^{\star}$

Target distribution

$$p^*(x_0) = \int \prod_{t=T+1}^1 p_t^*(x_{t-1} | x_t) dx_{1:T}$$

■ IMPORTANCE SAMPLING

$$x_{t-1} \sim p_t^*(\cdot | x_t) = p_t^{\text{pre}}(\cdot | x_t) \underbrace{\frac{e^{v_{t-1}(\cdot)/\beta}}{\int p_t^{\text{pre}}(x | x_t) e^{v_{t-1}(x)/\beta} dx}}_{\text{importance weight}} \approx \sum_{m=1}^M \delta_{x_{t-1}^{(m)}} \frac{w_{t-1}^{(m)}}{\sum_{j=1}^M w_{t-1}^{(j)}}$$

★ M samples $\{x_{t-1}^{(m)}\}_{m=1}^M \sim p_t^{\text{pre}}(\cdot | x_t)$

★ optimal value function $w_{t-1}^{(m)} = e^{\hat{v}_{t-1}(x_{t-1}^{(m)})/\beta}$

★ resample $x_{t-1} = x_{t-1}^{(\zeta_{t-1})}$ w/ $\zeta_{t-1} \sim \text{Categorical} \left(\left\{ \frac{w_{t-1}^{(m)}}{\sum_{j=1}^M w_{t-1}^{(j)}} \right\}_{m=1}^M \right)$

Cascaded sampling from $t = T + 1$ to $t = 1$

Summary

- RECAP

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 - Online training-free sampling